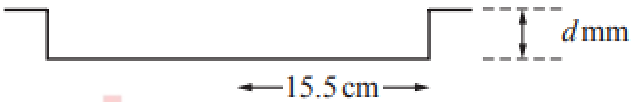
Mensuration

Worksheet 2e

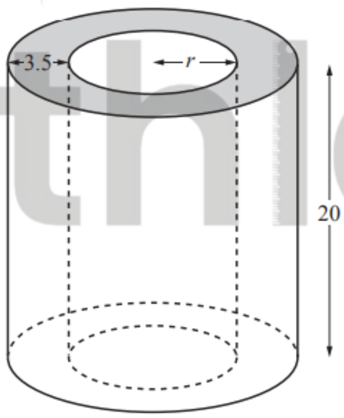
Cylinder

1. **S16/qp21/q7(b)(iii)**



To make a pizza, the baking tray is completely filled with dough to a depth of d mm. The open cylinder holds $500cm^3$ of dough. Calculate the depth of the dough, d mm, giving your answer correct to the nearest millimetre. [3]

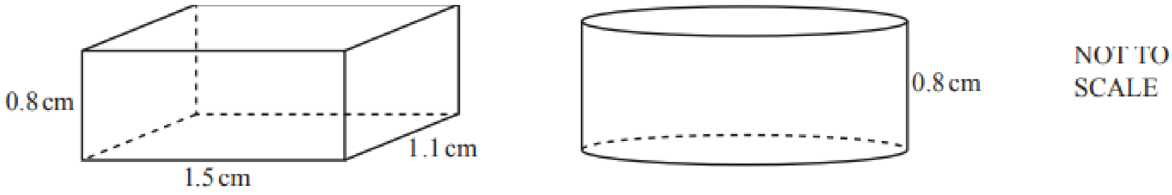
2. **S16/qp22/q11(a)(i)**



Solid I

Solid I is a cylinder with a small cylinder removed from its centre, as shown in the diagram. The height of each cylinder is 20cm and the radius of the small cylinder is r cm. The radius of the large cylinder is 3.5cm greater than the radius of the small cylinder. The volume of Solid I is $3000cm^3$.Calculate r. [4]

3. W16/qp42/q6(a)(ii)(iii)0580



The cuboid has length 1.5 cm, width 1.1 cm and height 0.8 cm. The cylinder has height 0.8 cm and the same volume as the cuboid.

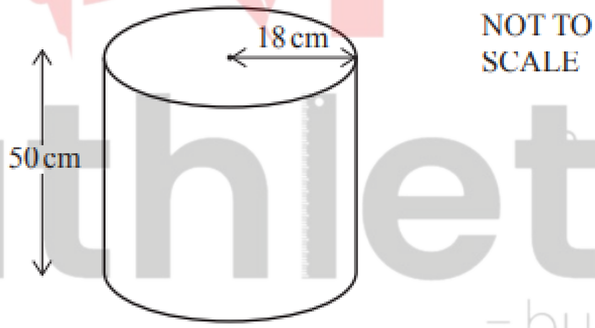
i. Calculate the volume of the cuboid. [2]

ii. Calculate the radius of the cylinder. [2]

iii. Calculate the difference between the surface areas of the two sweets. [5]

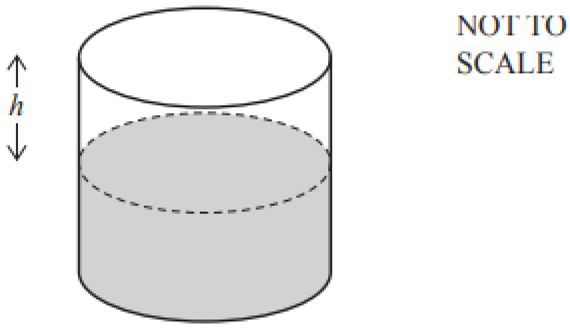
4. S17/qp41/q5(a)(i)(ii)0580

The diagram shows a cylindrical container used to serve coffee in a hotel. The container has a height of 50 cm and a radius of 18 cm.



i. Calculate the volume of the cylinder and show that it rounds to $50\ 900\ cm^3$, correct to 3 significant figures. [2]

ii. 30 litres of coffee are poured into the container. Work out the height, h, of the empty space in the container. [3]



5. S17/qp43/q4(b)0580

A cylinder of length 13 cm has volume $280\ cm^3$.

i. Calculate the radius of the cylinder. [3]

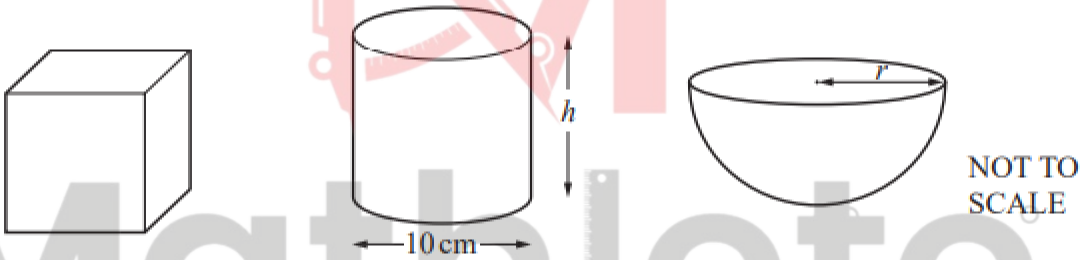
ii. The cylinder is placed in a box that is a cube of side 14 cm. Calculate the percentage of the volume of the box that is occupied by the cylinder. [3]

6. W17/qp21/q20 0580

a. A cylinder has height 20 cm. The area of the circular cross section is $74\ cm^2$. Work out the volume of this cylinder. [1]

b. Cylinder A is mathematically similar to cylinder B. The height of cylinder A is 10 cm and its surface area is $440\ cm^2$. The surface area of cylinder B is $3960\ cm^2$. Calculate the height of cylinder B. [3]

7. W17/qp43/q6(a)(i)0580



The diagrams show a cube, a cylinder and a hemisphere. The volume of each of these solids is $2000\ cm^3$. Work out the height, h, of the cylinder. [2]

8. S18/qp22/q15 0580

Find the volume of a cylinder of radius 5cm and height 8cm. Give the units of your answer. [3]

9. M19/qp22/q18 0580

A pipe is full of water. The cross-section of the pipe is a circle, radius 2.6cm. Water flows through the pipe into a tank at a speed of 12 centimetres per second. Calculate the number of litres that flow into the tank in one hour. [3]

Answer Key

1. S16/qp21/q7(b)(iii)

(iii)	7	3	M1 for $\pi 15.5^2 d = 500$ A1 for 0.66 to 0.663

2. S16/qp22/q11(a)(i)

11 (a) (i)	5.06 to 5.08	4	**B1** for $r + 3.5$ seen **B1** for $\pi(r + 3.5)^2 - \pi r^2$ or $20\pi(r + 3.5)^2 - 20\pi r^2$ **B1** for $20\pi(r + 3.5)^2 - 20\pi r^2 = 3000$ or better

3. W16/qp42/q6(a)(ii)(iii)0580

6 (a) (i)	1.32	2	**M1** for $0.8 \times 1.5 \times 1.1$
(ii)	0.725 or 0.7246 to 0.7247…	2	**M1** for $\pi r^2 \times 0.8 = \textit{their}$**(a)(i)** or $\pi r^2 = 1.5 \times 1.1$ oe
(iii)	0.513 to 0.518 nfww	5	**M1** for $2(1.5 \times 1.1 + 1.5 \times 0.8 + 1.1 \times 0.8)$ **M1** for $[2 \times]\ \pi \times (\textit{their}$ **(a)(ii)**$)^2$ **M2** for $\pi \times 2 \times (\textit{their}$ **(a)(ii)**$) \times 0.8$ or **M1** for $\pi \times 2 \times (\textit{their}$ **(a)(ii)**$)$

4. S17/qp41/q5(a)(i)(ii)0580

5(a)(i)	50890 or 50893 to 50900.4	2	**M1** for $\pi \times 18^2 \times 50$
5(a)(ii)	20.5 or 20.52 to 20.534	3	**B2** for answer 29.5 or 29.46 to 29.48 OR **M2** for $(50900 - 30000) \div (\pi \times 18^2)$ oe or **M1** for (figs 50.9 –figs 30) $\div$ ($\pi \times$ figs18^2) or **M1** for $(50900 - 30000) = (\pi \times 18^2)h$ oe OR **alternative method** **M2** for $50 - \frac{30000}{\pi \times 18^2}$ oe **M1** for figs $30 = \pi \times$ figs $18^2 \times (50 - h)$ oe or for $\frac{\text{figs}\,30}{\pi \times \text{figs}18^2}$ oe OR **alternative method** **M2** for $\frac{(50.9-30)}{50.9} \times 50$ oe or **M1** for $\frac{(50.9-30)}{50.9}$ or $\frac{30}{50.9} \times 50$ oe or **M1** for $\frac{(\text{figs } 50.9 - \text{figs } 30)}{\text{figs } 50.9} \times 50$ oe

5. S17/qp43/q4(b)0580

4(b)(i)	2.62 or 2.618…	3	**M2** for $[r^2 =]\ \frac{280}{13\pi}$ oe or **M1** for $280 = \pi \times r^2 \times 13$
4(b)(ii)	10.2 or 10.20… or $10\frac{10}{49}$	3	**M2** for $\frac{280}{14^3}$ [×100] oe or **B1** for 2 744 or 14^3 seen

6. W17/qp21/q20 0580

20(a)	1480	1	
20(b)	30	3	**M2** for $10 \times \sqrt{\frac{3960}{440}}$ or $10 \div \sqrt{\frac{440}{3960}}$ or **M1** for $\sqrt{\frac{3960}{440}}$ or $\sqrt{\frac{440}{3960}}$ or $\left(\frac{h}{10}\right)^2 = \frac{3960}{440}$ oe

7. W17/qp43/q6(a)(i)0580

6(a)(i)	25.5 or 25.46…	2	**M1** for $\pi \times 5^2 \times h = 2000$ oe

8. S18/qp22/q15 0580

15	628 or 628.3 to 628.4 cm^3	3	**B2** for 628 or 628.3 to 628.4 or **M1** for $5^2 \times 8 \times \pi$ **B1** for cm^3

9. M19/qp22/q18 0580

18	917 or 918 or 917.4 to 917.6	3	**M2** for $\pi \times 2.6^2 \times 12 \times 60 \times 60 \div 1000$ or **M1** for $\pi \times 2.6^2$ isw or $12 \times 60 \times 60 \div 1000$ isw If 0 scored **SC1** for figs 917 to 918